Table 1: Method comparison on CORE4D-G1 (50 paper sequences), mean over cases. **SPIDER-CEM** (ours) vs **OmniRetarget** (our upstream kinematic input), **SBTO**, and **GMR**. Physical contact / penetration are frame fractions at 3 mm and 10 mm bands from the same MuJoCo contact detector; max penetration is the deepest hand-object interpenetration. ↑/↓: higher/lower is better; best per row in bold, second best underlined. OmniRetarget (kinematic, no collision) penetrates heavily at modest clean contact; SPIDER-CEM's physics optimisation raises contact@3mm to 59.1% while cutting penetration on every threshold. GMR is object-unaware: near-zero clean contact (contact@3mm $\approx 0.4\%$) yet the deepest max penetration (99 mm) – read its penetration jointly with contact, never alone.

Metric	SPIDER-CEM	OmniRetarget	SBTO	GMR
Phys. contact@3mm (%) ↑	**59.08**	9.40	<u>19.12</u>	0.39
Phys. contact@10mm (%) ↑	**80.49**	49.09	<u>65.09</u>	1.33
Phys. penetration@3mm (%) ↓	**17.47**	53.47	48.71	<u>34.38</u>
Phys. penetration@10mm (%) ↓	**1.74**	25.68	<u>13.36</u>	33.44
Max phys penetration (mm) ↓	**12.40**	<u>21.61</u>	25.36	99.12
Obj speed max (m/s) ↓	**1.29**	2.77	<u>1.76</u>	1.82
Foot slip max (m) ↓	**0.87**	1.33	1.20	<u>0.97</u>